

AmirHossein Fallah Zarrinkar

(Amir Fallah)

Expert Project Manager/Physics Team Lead @ Mercor AI

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EDUCATION

MSc in Physics 2022-2024, CA
Université de Montréal (UdeM)
Grade: 4/4.3

- Thesis: “Charged operators in Gross-Neveu family quantum critical points using large-N expansion” in [Fr](#), [En](#)
- Research Advisor: Prof. [William Witzak-Krempa](#): Canada Research Chair in Quantum Phase Transitions

B.S. in Physics, “Summa cum laude” 2016-2021, IR
U. of Guilan (UofG), Sharif U. of Tech., and Institute for Research in Fundamental Sciences (IPM)
Grade: 18.78/20 (4.00/4.00¹), Rank 1 among 75 students of the faculty of Physics

- Thesis: “The entropy of Hawking radiation, quantum information in black hole evaporation”
- Research Advisor: [M. Reza Mohammadi Mozaffar](#)

B.S. in Mathematics, “Summa cum laude” 2016-2021, IR
University of Guilan (UofG) and IPM
Grade: 18.32/20 (3.91/4.00), Rank 1 among 55 students of the faculty of Mathematics

INTERESTS

- **AI for science, and science for AI, Quantum computing, Quantum information, Deep learning, World models, V-JEPA, VLM, RL, Geometric Deep Learning.** I am always eager to learn and research.

SKILLS

- **Python, Numpy, Matplotlib, XGBoost, Pandas, sklearn, Pytorch, Mathematica, Matlab, Fortran, Machine Learning Specialization, IBM Data Science Professional Certificate, L^AT_EX.**
- **Math: Group theory and symmetries, Lie Algebra, Topology and geometry of manifolds.**
- **Physics: Quantum Field Theory, Condensed Matter Physics.**

SCHOLARSHIPS AND AWARDS

- **Fast-track to PhD scholarship**, Université de Montréal, 10000 CAD 2022
- **Physics department entrance scholarship**, Université de Montréal , 5000 CAD 2022
- **Abrishamchian Prize**, Prize of the **best bachelor student of Guilan university**, 1000 CAD 2019
- **Ranked 5th** in national university students physics Olympiad 2019
- **Ranked 1st** in north university student physics Olympiad 2019
- **Prize of the best physics student of Guilan university** 2018
- **Semifinalist** in the National Physics Olympiad, Young Scholars Club 2014
- **Iran Science and Technology Park Research Fellowship** for building a stair climbing robot (1000\$) 2013

¹The GPA out of 4 is calculated based on the information provided by the [World Education Services](#).

- Ranked 1st in Sharif Nadcup* Robotics tournament in MicroMouse 2013
*The greatest robotics competition for Iranian students

EXPERIENCES AND EXTRACURRICULAR ACTIVITIES

- **Expert Project Manager (EPM) & Physics Team Lead & Math Senior Reviewer @ Mercor AI** Since–2025
Collaborating to develop LLMs (including models from OpenAI, Google DeepMind, Anthropic, and Meta) to solve complex mathematics and physics problems.
Promoted to EPM/Physics Team Lead/Math Senior Reviewer.
- **AI Assessor and Senior Reviewer @ Scale AI, Outlier** Since–2024
Collaborating to develop LLMs to solve complex mathematics and physics problems. Promoted to Senior Reviewer and ORACLE based on consistent high-quality work and performance metrics.
- **Teaching Assistant** UdeM, UofG 2017–2024
Helping students to perform their experiments Teaching and helping students to utilize Python (NumPy, Matplotlib) to analyze their results, Grading the students.
- **Participation** Bootstrapping Nature: Non-perturbative Approaches to Critical Phenomena 2022
at GGI INFN, Florence, Italy
- **Co-Founder at Sinoacademy** 2019- 2020
A Persian open online course provider start-up,
We ran a summer school on physics in machine learning.
• I with my co-founders organized the school with 7 teachers for more than 50 students.
- **Participation And talk** in weekly journal club of Professor Hessamadin Arfaei Since 2020
High Energy and Gravitational Physics, Sharif University of Technology
Talk On Entropy of Hawking Radiation, [Download the presentation file](#)
- **Invited speaker of physics day** at University of Guilan 2019
A talk on Quantum Foundations
- **Head of Olympiad department** at Mirza Kouchak Khan 2 high-school (NODET²) 2017–2018
*In this year, the students got the **best results** ever in the history of the high school*

RELATED COURSES

Audited: [Ongoing: Deep Learning](#) at NYU, Half of IFT 6135 – Representation Learning in Fall 2024. McGill COMP 451 Fall 2023: Fundamentals of Machine Learning

Courses: Advanced Programming, Linear optimization, Linear Algebra, Numerical Linear Algebra, Probability, Statistics, Manifolds, Advanced Topology, Algebra, Group theory and symmetries, Measure Theory, Quantum Field Theory, Quantum Condensed Matter, Introduction to Quantum Computing, String Theory.

In total, I had more than 200 credits in Physics and Mathematics.

REFERENCES

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| <p>William Witczak-Krempa</p> <ul style="list-style-type: none"> • w.witczak-krempa@umontreal.ca
Associate Professor at UdeM | <p>Rufus Boyack</p> <ul style="list-style-type: none"> • Rufus.Boyack@dartmouth.edu, Google Scholar
Assistant Prof. at Dartmouth College |
| <p>Mohammad Reza Mohammadi Mozaffar</p> <ul style="list-style-type: none"> • mmohammadi@guilan.ac.ir, iNSPIRE
Associate Prof. at UofG Resident Researcher at IPM | <p>Hessamaddin Arfaei</p> <ul style="list-style-type: none"> • arfaei@sharif.edu, iNSPIRE
Professor at SUT, Founding Fellow of IPM (Retired) |

²National Organization for Development of Exceptional Talents